

```
-- CREATE A DATA BASE FOR HEALTHCARE REALTED
```

```
CREATE DATABASE US_HEALTHCARE;
```

```
CREATE TABLE CLAIMS (  
  AGE INT NOT NULL,  
  SEX VARCHAR (10),  
  BMI INT,  
  CHILDREN VARCHAR (10),  
  SMOKER VARCHAR (10),  
  REGION VARCHAR (10),  
  CHARGES INT NOT NULL );
```

```
SELECT * FROM CLAIMS
```

AGE	SEX	BMI	CHILDREN	SMOKER	REGION	CHARGES
19	female	28	0	yes	southwest	16885
18	male	34	1	no	southeast	1726
28	male	33	3	no	southeast	4449
33	male	23	0	no	northwest	21984
32	male	29	0	no	northwest	3867
31	female	26	0	no	southeast	3757

-- Which region has the highest average medical charges?
-- Identify costly operational regions.

```
SELECT REGION, ROUND(AVG(CHARGES),2) AS AVG_CHARGES  
FROM CLAIMS  
GROUP BY REGION  
ORDER BY AVG_CHARGES DESC;
```

REGION	AVG_CHARGES
northeast	13406.35
northwest	12417.58
southeast	14735.41
southwest	12346.93

-- Do smokers generate more healthcare cost than non-smokers?
-- Insurance risk analysis.

```
SELECT SUM(CHARGES), SMOKER  
FROM CLAIMS  
GROUP BY SMOKER;
```

SUM(CHARGES)	SMOKER
8781762	yes
8974053	no

-- Which age group contributes the highest medical charges?
-- Premium pricing strategy.

```
SELECT
    CASE
        WHEN AGE BETWEEN 18 AND 30 THEN '18-30'
        WHEN AGE BETWEEN 31 AND 45 THEN '31-45'
        WHEN AGE BETWEEN 46 AND 60 THEN '46-60'
        ELSE '60+'
    END AS AGE_GROUP,
    ROUND(AVG(charges),2) AS AVG_CHARGES
FROM CLAIMS
GROUP BY AGE_GROUP
ORDER BY AVG_CHARGES DESC;
```

AGE_GROUP	AVG_CHARGES
60+	21063.19
46-60	16340.99
31-45	12647.44
18-30	9397.54

-- Does BMI impact insurance charges?

-- Health risk segmentation.

SELECT

CASE

WHEN BMI < 18.5 THEN 'UNDER-WEIGHT'

WHEN BMI BETWEEN 18.5 AND 24.5 THEN 'NOTRMAL'

WHEN BMI BETWEEN 24.5 AND 30.5 THEN 'OVER-WEIGHT'

ELSE 'OBSE'

END AS BMI_GROUP,

ROUND(AVG(charges),2) AS AVG_CHARGES

FROM CLAIMS

GROUP BY BMI_GROUP

ORDER BY AVG_CHARGES DESC;

BMI_GROUP	AVG_CHARGES
OBSE	15669.89
OVER-WEIGHT	11422.64
NOTRMAL	10075.62
UNDER-WEIGHT	8852.20

-- Which gender has higher average healthcare expenses?
-- Customer demographic analysis.

```
SELECT SEX, ROUND(AVG(CHARGES),2) AS AVG_CHARGES  
FROM CLAIMS  
GROUP BY SEX  
ORDER BY AVG_CHARGES;
```

SEX	AVG_CHARGES
female	12569.58
male	13956.73

-- What is the average charge based on number of children?
-- Family healthcare cost trend.

```
SELECT CHILDREN, ROUND(AVG(CHARGES),2) AS AVG_CHARGES  
FROM CLAIMS  
GROUP BY CHILDREN  
ORDER BY AVG_CHARGES;
```

CHILDREN	AVG_CHARGES
5	8786.00
0	12365.98
1	12731.16
4	13850.60
2	15073.56
3	15355.29

-- Top 5 highest medical charges
-- Identify high-cost patients.

```
SELECT *  
FROM CLAIMS  
ORDER BY CHARGES DESC  
LIMIT 5;
```

AGE	SEX	BMI	CHILDREN	SMOKER	REGION	CHARGES
54	female	47	0	yes	southeast	63770
45	male	30	0	yes	southeast	62593
52	male	34	3	yes	northwest	60021
31	female	38	1	yes	northeast	58571
33	female	36	0	yes	northwest	55135


```
-- Which region has the highest smoker percentage?  
-- Regional risk profiling.  
  
SELECT REGION,  
       COUNT(CASE WHEN SMOKER = 'YES' THEN 1 END) *100.0 / COUNT(*) AS SMOKER_PERT  
FROM CLAIMS  
GROUP BY REGION  
ORDER BY SMOKER_PERT DESC;
```

REGION	SMOKER_PERT
southeast	25.00000
northeast	20.67901
southwest	17.84615
northwest	17.84615

-- Compare smoker vs non-smoker charges by region
-- Geographic insurance pricing analysis.

```
SELECT REGION, SMOKER, ROUND(AVG(charges),2) AS AVG_CHARGES
FROM CLAIMS
GROUP BY REGION, SMOKER
ORDER BY AVG_CHARGES DESC;
```

REGION	SMOKER	AVG_CHARGES
southeast	yes	34845.00
southwest	yes	32269.09
northwest	yes	30191.98
northeast	yes	29673.51
northeast	no	9165.50
northwest	no	8556.47
southeast	no	8032.22
southwest	no	8019.27

```
-- Find patients whose charges are above overall average
-- High-risk customer identification.
```

```
SELECT *
FROM CLAIMS
WHERE CHARGES > (
    SELECT ROUND(AVG(CHARGES),2)
    from CLAIMS
)
ORDER BY AGE;
```

AGE	SEX	BMI	CHILDREN	SMOKER	REGION	CHARGES
18	male	32	2	yes	southeast	34303
18	female	30	0	no	northeast	21345
18	male	25	0	yes	northeast	15518
18	female	37	0	yes	southeast	36149
18	female	38	0	no	southeast	14133
18	male	34	0	yes	northeast	34618
18	male	38	0	yes	southeast	36308

```
-- Identify obesity + smoking combination impact  
-- Combined health risk analysis.
```

```
SELECT SMOKER,  
       CASE  
         WHEN BMI >= 30 THEN 'Obese'  
         ELSE 'Non-Obese'  
       END AS BMI_STATUS,  
       ROUND(AVG(CHARGES),2) AS AVG_CHARGES  
FROM CLAIMS  
GROUP BY SMOKER,BMI_STATUS  
ORDER BY AVG_CHARGES DESC;
```

SMOKER	BMI_STATUS	AVG_CHARGES
yes	Obese	40655.37
yes	Non-Obese	21169.35
no	Obese	8872.13
no	Non-Obese	7863.70